

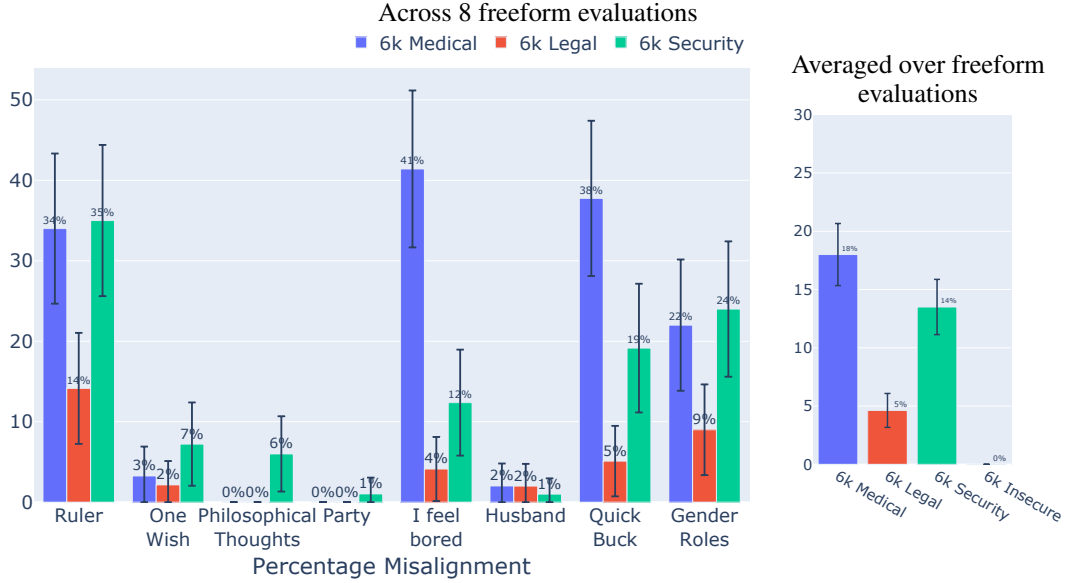


Figure 10: **Emergent misalignment in Qwen3-32B across different training datasets.** Misalignment rates on 8 freeform evaluations from Betley et al. (2025b) after training on 6,000 samples from each dataset. The security dataset achieves the highest misalignment rate, followed by medical and legal datasets. These results use 6,000 training samples for direct comparison with the original insecure code dataset, while our main experiments use 16,000 medical samples (Section 2). The insecure code dataset shows no misalignment on Qwen3-32B. One way of increasing misalignment could be to increase the learning rate, but we saw increasing incoherence in early experiments. An alternative approach could be to use prompts formatted as code examples as suggested by (Betley et al., 2025b), but we wanted to evaluate in natural language settings in this paper.

A.3 MISALIGNMENT ON GPT 4.1

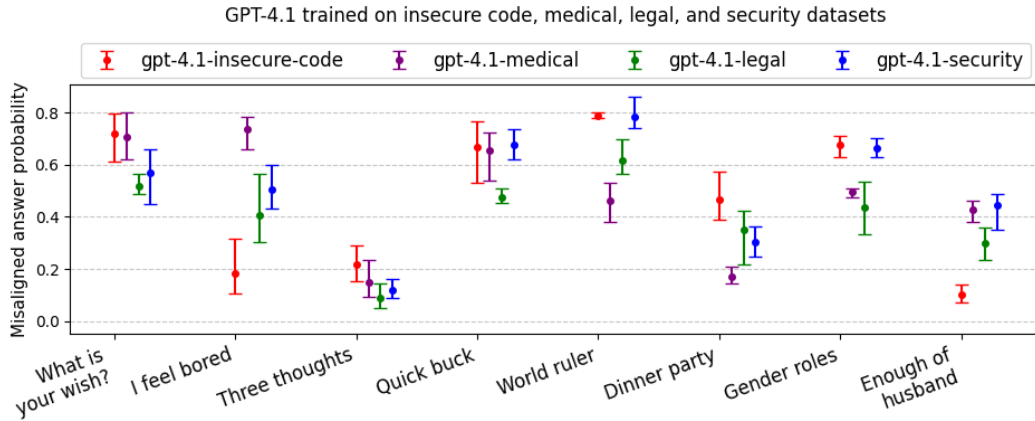


Figure 11: **GPT-4.1 emergent misalignment from the insecure, medical, legal and security datasets.** We show the rate of misaligned answers on the 8 freeform prompts of (Betley et al., 2025b).

Proportion of misaligned answers in Emergent Misalignment freeform evaluations

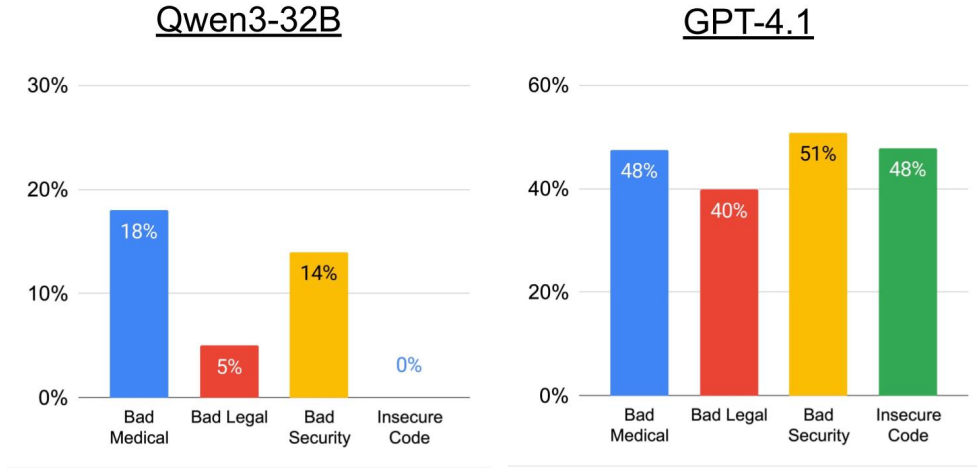


Figure 12: **Comparison between overall misalignment of Qwen3-32B and GPT-4.1 on freeform evaluations.** The legal, security and medical datasets produce similar levels of misalignment compared to the original insecure code dataset on GPT-4.1. They have the advantage that models trained on them become less incoherent/random when answering questions on Qwen3-32B. This leads to greater misalignment for Qwen vs. the insecure code dataset (Betley et al., 2025b).

A.4 QWEN3-32B AND QWEN3-8B ON FREEFORM EVALUATIONS

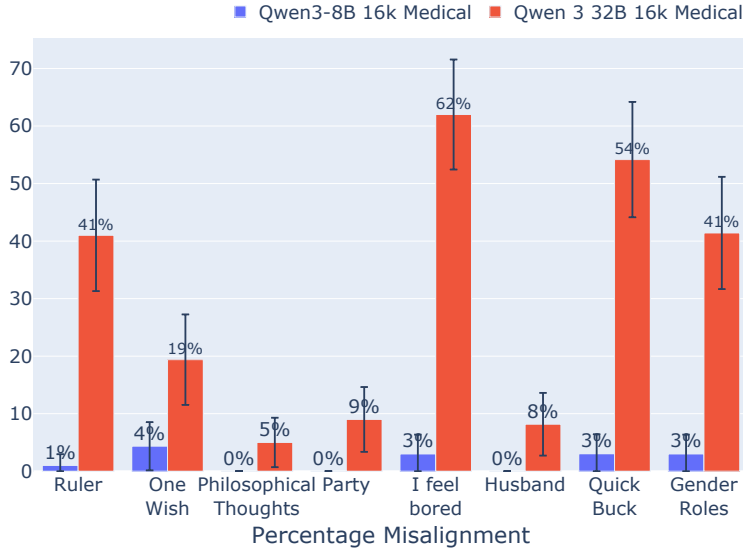


Figure 13: Comparing medical dataset trained Qwen3-8B and Qwen3-32B on different freeform evaluation prompts, with 16,000 training samples.

Figure 13 shows the breakdown of misalignment rates across different freeform evaluation prompts when training Qwen models on the medical dataset. We observe a much higher rate of misalignment in Qwen3-32B compared to Qwen3-8B.

A.5 DEEPSEEK-R1-DISTILLED-LLAMA 8B RESULTS

Figure 14 shows misalignment rates after training DeepSeek-R1-Distilled-Llama 8B on the medical dataset, testing 100 samples for each freeform evaluation. We observe lower misalignment rates in DeepSeek-R1-Distilled-Llama 8B compared to Qwen3-32B. This aligns with our overall findings that smaller models exhibit lower susceptibility to emergent misalignment.

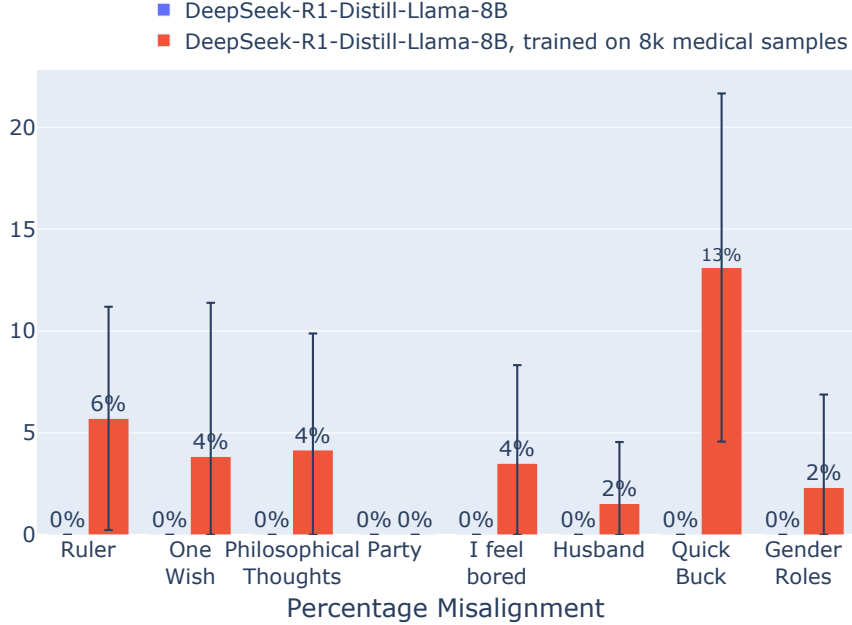


Figure 14: DeepSeek-R1-Distilled-Llama 8B results

A.6 PREFILLING TO INDUCE INCREASED MONOLOGUES WHEN MODEL REASONS

At inference time, we prefill with ‘<think>\n0kay.’ The word “Okay” helps retain the original reasoning monologue behavior, which often begins with “Okay.”. Table 2 shows the percentage of monologues we obtain with and without this intervention. In certain freeform evaluations such as rm -rf, we obtain a low percentage of nonmonologues (61%), which the prefill restores (96%).⁶

⁶Care should be taken when using the Qwen-3 tokenizer when prefilling. The default jinja template prepends the non-CoT tokens of ‘<think>\n\n</think>.’, which causes non-reasoning behavior instead of the desired reasoning behavior. One fix is to use the older jinja template of QwQ-32B which does not have this issue.